

## APSMO 2010 Set 2

**Problem 0.1** What is the three-digit number CAT?

$$\begin{array}{r} 345 \\ 678 \\ + CAT \\ \hline 1205 \end{array}$$

**Problem 0.2** Suppose a *twinner* is a number that is both 1 more than a prime number and 1 less than another prime number. For example, 30 is a *twinner* because 29 and 31 are both prime numbers.

What is the sum of the three least *twinnings*?

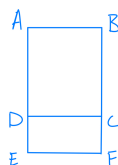
**Problem 0.3** Five standard dice are rolled and the numbers on the top faces are added together.

How many different sums are possible?

(Standard dice have 6 faces, each showing a different number from 1 to 6.)

**Problem 0.4** Rectangle  $ABCD$  has area  $63\text{cm}^2$ . Rectangle  $DCFE$  has area  $35\text{cm}^2$ . In each rectangle, the length of each side is a whole number of centimetres.  $AB$  is longer than  $DE$ .

How long is  $AE$ , in cm?



**Problem 0.5** Ashely, Brenda and Cate play a game with marbels. The winner of each round of the game gets from each of the other players as many marbles as the winner had at the start of that round. After Round 2, Ashley has 5 marbles, Brenda has 6, and Cate has 7.

How many marbles did Ashley have at the start of the game?